

PHISHING WEBSITE IMPERSONATION: COMPARATIVE ANALYSIS OF DETECTION AND TARGET RECOGNITION METHODS

Presented By : Marcin Jarczewski, Piotr Białczak, Wojciech Mazurczyk

Reg No. : ASI22CA035 / CERT Polska & WUT

Guide : Asst. Prof. Asha Rose Thomas

Semester : S7, CSE(AI)

CONTENTS

- **Cybersecurity Challenge** & Phishing Escalation
- **Limitations of Traditional** Heuristics & Code Analysis
- **The Visual Analysis Solution** & Dual Research Objectives
- **Key Technologies:** Deep Learning vs. Perceptual Hashing
- **Evaluated Method 1:** VisualPhishNet (Triplet CNN Layout)
- **Evaluated Method 2:** Phishpedia (Faster R-CNN + Siamese)
- **Proposed Baseline Method:** Perceptual Hashing (pHash + FAISS)
- **Experimental Framework** & Dataset Standardization
- **Thresholding Strategy:** Equal Error Rate (EER) Optimization
- **Performance Analysis:** Binary Classification & Feature Collapse
- **Performance Analysis:** Multiclass Target Brand Recognition
- **Hybrid Operational Pipeline** & Real-World Applications
- **Industry Impact & Future Scope**
- **Conclusion** & Key Takeaways

- **The Rapid Digitization of Global Commerce:** Over 72% of enterprises maintain online portals, with 27% of global sales executed digitally.
- **Escalating Scale of Cybercrime:** Observed phishing attacks surpassed 1.13 Million in Q2 2025 (rising from 1.00M in Q1 2025).
- **Primary Attack Vectors & Delivery:** Fraudulent websites, email campaigns, SMS (smishing), and voice calls (vishing).
- **Key Problems for Security Teams:**
 - o **High Volume & Sophistication:** Attackers deploy fast-flux networks and automated site generators.
 - o **High Economic Impact:** Credential theft, financial fraud, and severe organizational brand damage.
 - o **Passive Defense Limits:** Static blocklists fail against zero-day phishing infrastructure.

LIMITATIONS OF TRADITIONAL HEURISTICS

- **The Obfuscation Paradox:** Attackers heavily scramble source code while maintaining identical visual rendering for victims.
- **Weaknesses of Legacy Detection:**
 - **URL & Text Heuristics:** Easily bypassed via dynamic subdomains, FreeURL manipulation, and IP redirection.
 - **HTML Source Code Analysis:** Vulnerable to JavaScript obfuscation, DOM hiding, and dynamic content fetching.
 - **Early Computer Vision (EMD/SIFT/DAISY):** Computationally prohibitive for real-time traffic and lack generalization.
- **The Imperative for Visual Analysis:** Regardless of code obfuscation, the visual representation must mimic the trusted brand to deceive users.

THE CORE IDEA: PROPOSED SOLUTION

- **Visual Phishing Detection & Target Recognition Pipeline:** Treating webpage screenshots as the ground-truth data source for automated security triage.
- **Dual Research Objectives:**
 - **Binary Phishing Detection:** Determine whether a site possesses malicious intent (Phishing vs. Benign).
 - **Multiclass Target Recognition:** Identify the exact brand/institution being impersonated to enable specific incident response.
- **The Concept in 3 Steps:**
 - **INPUT:** Webpage screenshot collected by operational crawler / CERT analyst.
 - **PROCESSING:** Standardized microservices pipeline combining feature extraction & vector matching.
 - **OUTPUT:** Binary threat verdict + exact impersonated brand identification.

KEY TECHNOLOGY 1: PERCEPTUAL HASHING & FAISS

- **The Proposed Lightweight Baseline:** Perceptual Hashing (pHash) combined with FAISS similarity search.
- **Locality-Sensitive Hashing (LSH) Concept:** Unlike cryptographic hashes (MD5/SHA256) where minor edits alter the output, LSH maps visually similar images to close hash buckets.
- **Discrete Cosine Transform (DCT) pHash:**
 - **Frequency Conversion:** Transforms screenshots into frequency domains, preserving structural layout features while stripping noise.
 - **Vector Hashing (pHashF):** Generates real-valued vectors for dense similarity evaluation.
- **FAISS Library Integration:** Enables high-speed Euclidean and Hamming distance vector indexing across millions of reference targets.

KEY TECHNOLOGY 2: DEEP LEARNING VISUAL MODELS



- **State-of-the-Art Deep Learning Approaches:** Automated neural feature extraction replacing handcrafted vision descriptors.
- **Two Distinct Design Philosophies:**
 - **Holistic Layout Analysis (VisualPhishNet):** Evaluates the overall visual 'feel', spatial hierarchy, and layout composition of the entire webpage.
 - **Targeted Object Detection (Phishpedia):** Focuses specifically on high-precision brand logo extraction and identity matching.
- **The Critical Research Question:** Do complex deep neural networks consistently outperform lightweight baseline algorithms across diverse, real-world datasets?

DEEP DIVE: VISUAL SIMILARITY VS OBJECT DETECTION

- **Holistic Layout Matching vs. Logo Object Detection:**
- **Holistic Approach (Layout-Based):**
 - o **Premise:** Login pages and homepages of a brand share consistent visual styling.
 - o **Mechanism:** Convolutional Neural Networks (CNNs) encode the full screenshot into a dense embedding space.
- **Targeted Approach (Logo-Based):**
 - o **Premise:** Phishing sites must display authentic brand logos to gain victim trust.
 - o **Mechanism:** Bounding box proposal models detect logo regions and compare them against protected reference databases.

MODEL ARCHITECTURE 1: PHISHPEDIA

- **Phishpedia Architecture (Lin et al.):** A high-precision two-stage visual brand recognition pipeline.
- **Stage 1: Logo Region Proposal:** Utilizes a Faster R-CNN object detection model fine-tuned on the Logo-2K+ dataset to crop potential logo bounding boxes.
- **Stage 2: Siamese Identity Matching:** Passes cropped regions into a Siamese Network to compute distance against a protected brand reference database.
- **Key Design Choice:** Replaces standard triplet loss with classification-based fine-tuning to effectively distinguish fine-grained brand variants (e.g., 'Adobe' vs. 'Adobe AIR').

MODEL ARCHITECTURE 2: VISUALPHISHNET

- **VisualPhishNet Architecture (Abdelnabi et al.):** A holistic webpage layout representation network.
- **Core Backbone:** Uses a VGG-16 Convolutional Neural Network trained with Triplet Loss.
- **Triplet Loss Mechanics:**
 - **Inputs:** Anchor image (reference site), Positive sample (same brand page), Negative sample (different brand page).
 - **Objective:** Minimizes Euclidean distance between Anchor and Positive while maximizing distance to Negative by a defined margin.
- **Hard Negative Mining:** Selects visually confusing non-brand pages during training to refine decision boundaries.

WORKFLOW STEP 1: DATASET CURATION & STANDARDIZATION



- **Addressing Dataset Fragmentation:** Evaluated across three comprehensive datasets to eliminate single-benchmark bias.
- **The Three Evaluated Datasets:**
 - o **CERT Polska Dataset:** 15,049 real-world, analyst-verified URLs & screenshots (36 target brands, unaugmented).
 - o **Phishpedia (PP) Dataset:** 14,500 phishing / 1,542 benign screenshots (56 targets, augmented).
 - o **VisualPhishNet (VP) Dataset:** 4,644 phishing / 8,835 benign screenshots (144 targets, augmented).
- **Stratified Data Splitting:** 60% Training, 20% Validation, 20% Testing split applied consistently across all target classes.

WORKFLOW STEP 2: EXPERIMENTAL MICROSERVICES FRAMEWORK

- **Containerized Microservices Architecture:** Engineered to resolve 'dependency hell' across conflicting deep learning library versions.
- **Framework Layer Breakdown:**
 - **API Gateway (FastAPI):** Acts as central orchestrator, validating Base64 screenshot payloads via Pydantic.
 - **Isolated Model Containers:** Dedicated Docker containers hosting VisualPhishNet, Phishpedia, and Baseline models independently.
 - **Persistence Layer (SQLite & SQLAlchemy):** Captures binary predictions, target attribution labels, and confidence metrics for post-hoc analysis.
- **Open Source Release:** Shared with the cybersecurity community on GitHub for reproducible benchmarking.

WORKFLOW STEP 3: THRESHOLDING & EER OPTIMIZATION

- **Distance-Based Decision Boundaries:** Classification relies on embedding distance to nearest protected target.
- **Two-Stage Threshold Optimization Strategy:**
 - **Coarse Search:** Evaluates distance thresholds from 0 to maximum validation distance in steps of 10.
 - **Fine-Grained Search:** Refines search within Mean $\pm$ 1 Std Dev in single-unit increments.
- **Equal Error Rate (EER) Point:** Identifies the exact threshold where False Positive Rate (FPR) equals False Negative Rate (FNR).
- **Optimal Threshold Findings:** CERT Polska = 8.00 | Phishpedia (PP) = 3.00 | VisualPhishNet (VP) = 50.00.

WORKFLOW STEP 4: BINARY CLASSIFICATION & FEATURE COLLAPSE

- **Binary Detection Results (Phishing vs. Benign):**

- o **Baseline (pHash) Superiority:** Achieved top binary performance across all datasets (PP F1 = 0.9539, VP ROC AUC = 0.8201, VP MCC = 0.6294).

- o **Phishpedia Binary Performance:** Strong on PP dataset (F1 = 0.9062), but struggled on CERT Polska (F1 = 0.1598) due to strict logo dependence.

- **VisualPhishNet Failure & Feature Collapse:**

- o **Low Metrics:** Achieved F1 of only 0.1673 on PP dataset and negative MCC values (-0.6160).

- o **Root Cause:** Overfitted to synthetic data artifacts; 85% of detected phishing samples collapsed into just 3 target clusters (Adobe, Absa, Paschoalotto).

WORKFLOW STEP 5: MULTICLASS BRAND ATTRIBUTION

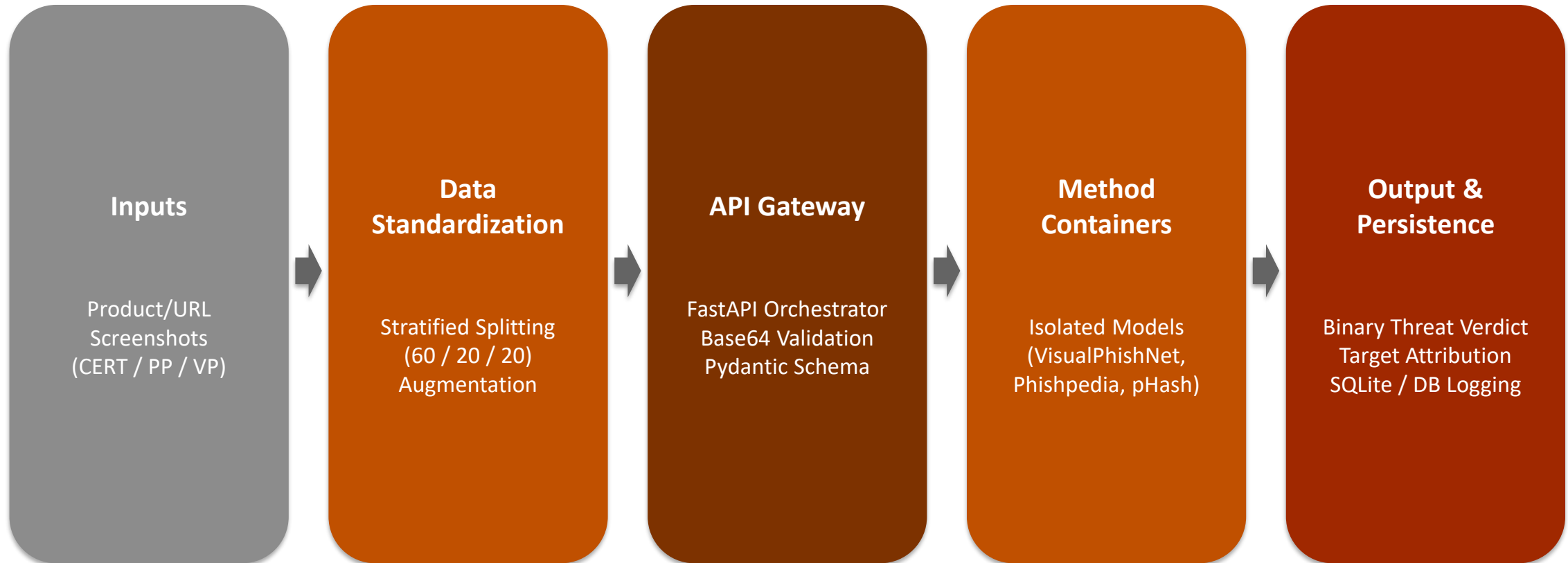
- **Multiclass Impersonation Target Identification:** Evaluating exact brand attribution among 36 to 144 target classes.
- **Phishpedia Dominance in Target Recognition:**
 - **High Identification Rate:** Maintained >0.90 Identification Rate across ALL datasets (CERT = 0.9845, VP = 0.9270, PP = 0.9154).
 - **Robustness:** Logo region extraction effectively handles brand variants regardless of site layout changes.
- **Baseline Multiclass Performance:** Achieved solid F1 Micro (VP = 0.7009) but lower Identification Rate (0.2554 - 0.5687), proving better suited for binary filtering than exact brand attribution.

EVALUATING THE OUTPUT: METRICS & BENCHMARKS



- **Comprehensive Evaluation Metrics:**
- **Binary Metrics:**
 - **F1 Score & MCC:** Matthews Correlation Coefficient (MCC) accounts for heavy class imbalance across all 4 confusion matrix quadrants.
 - **ROC AUC:** Measures class separability independent of fixed decision cut-off thresholds.
- **Multiclass Metrics:**
 - **Macro F1 vs. Micro F1:** Macro F1 treats all brands equally (protecting niche targets); Micro F1 measures global volume accuracy.
- **Identification Rate (Id / Rep_TP):** Measures recall of correct brand attribution strictly on correctly detected phishing pages.

SYSTEM WORKFLOW & ARCHITECTURE



EXPERIMENTAL RESULTS & COMPARATIVE TABLE

Dataset	Method	F1 Micro	F1 Macro	MCC	Identification Rate
CERT Polska	VisualPhishNet	0.3654	0.1481	0.0733	0.7093
CERT Polska	Phishpedia	0.5482	0.2621	0.2013	0.9845
CERT Polska	Baseline (pHash)	0.5823	0.1670	0.3668	0.2554
VisualPhishNet (VP)	VisualPhishNet	0.3924	0.0047	0.0694	0.0037
VisualPhishNet (VP)	Phishpedia	0.3782	0.3073	0.2569	0.9270
VisualPhishNet (VP)	Baseline (pHash)	0.7009	0.4111	0.5089	0.5679
Phishpedia (PP)	VisualPhishNet	0.1003	0.0334	0.0111	1.0000
Phishpedia (PP)	Phishpedia	0.7691	0.2894	0.7384	0.9154
Phishpedia (PP)	Baseline (pHash)	0.5467	0.3591	0.4939	0.5687

REAL-WORLD APPLICATIONS & CERT CASE STUDY



- **Operational Deployment Scenarios:**

- o **National CSIRT / CERT Triage Pipeline:** Processing 15,000+ suspicious regional URLs daily with automated priority scoring.
 - o **Brand Protection & Anti-Phishing Monitoring:** Enables automated takedown notifications to impersonated institutions.
 - o **Browser Security Extensions:** Real-time client-side pHash verification before user credential submission.
- **CERT Polska Real-World Insights:** Unaugmented, analyst-verified threat data confirmed that baseline perceptual hashing provides superior stability over complex CNNs in real-time triage.

INDUSTRY IMPACT & OPERATIONAL EFFICIENCY



- **Shifting the Paradigm of Phishing Defense:**

- o **Computational Efficiency:** VisualPhishNet requires 8-10 hours training per dataset; Baseline pHash & Phishpedia complete training in ~20 minutes.

- o **Democratization:** Provides open-source containerized evaluation tools accessible to small SOC teams without heavy GPU clusters.

- o **Operational Agility:** Allows security teams to instantly update reference logo databases without full model retraining.

- **Analyst Fatigue Reduction:** Automates initial filtering, reserving human investigation for high-confidence brand matches.

FUTURE SCOPE AND POTENTIAL IMPROVEMENTS



- **What's Next for Visual Phishing Detection?**

- o **K-Fold Cross-Validation Framework:** Incorporating rigorous cross-validation to provide formal uncertainty quantification.
- o **Multimodal Deep Fusion:** Integrating visual pHash/Phishpedia features with DOM Graph Neural Networks (GNN) and LLM reasoning.
- o **Temporal Campaign Tracking:** Grouping detection results by time windows to model long-term phishing infrastructure evolution.
- o **Adversarial Robustness:** Enhancing resilience against adversarial visual perturbations (watermarking, subtle color shifts).

CONCLUSION

- **Summary of Key Findings:**

- o **Baseline (pHash) Superiority:** Perceptual hashing delivers the most robust binary classification across all benchmark datasets.
- o **Phishpedia Target Recognition:** Consistently achieves >0.90 Identification Rate, proving ideal for brand attribution.
- o **VisualPhishNet Vulnerability:** Suffers from feature collapse when trained on synthetic/augmented data distributions.

- **The Recommended Operational Hybrid Architecture:**

- o **Stage 1 (Rapid Filtering):** Use Perceptual Hashing (pHash + FAISS) for instant binary phishing detection.
- o **Stage 2 (Brand Identification):** Pass detected threats to Phishpedia for precise impersonation target recognition.

- **Jarczewski, M., Białczak, P., & Mazurczyk, W.** (2026). Phishing Website Impersonation: Comparative Analysis of Detection and Target Recognition Methods. MDPI Applied Sciences, 16(2), 640. DOI: 10.3390/app16020640
- **Lin, Y., Liu, R., Divakaran, D. M., et al.** (2021). Phishpedia: A Hybrid Deep Learning Based Approach to Visually Identify Phishing Webpages. In 30th USENIX Security Symposium (pp. 3793-3810).
- **Abdelnabi, S., Krombholz, K., & Fritz, M.** (2020). VisualPhishNet: Zero-Day Phishing Website Detection by Visual Similarity. In ACM Conference on Computer and Communications Security (CCS) (pp. 1633-1650).
- **Douze, M., Guzhva, A., Deng, C., et al.** (2025). The Faiss Library. arXiv preprint arXiv:2401.08281.
- **Framework Code Repository:** <https://github.com/Percival33/phish-target-recognition>

THANK YOU

Questions & Comments Welcome